Task 1: Basic Concepts

a. Define plaintext, ciphertext, encryption, and decryption.

- Plaintext: The original message in a readable format.
- Ciphertext: The encrypted message that cannot be easily understood without the correct key.
- Encryption: The process of converting plaintext into ciphertext using an algorithm and a key.
- Decryption: The process of converting ciphertext back into plaintext using the correct key.

b. Explain the difference between symmetric and asymmetric cryptography.

- Symmetric cryptography: Uses the same secret key for encryption and decryption. Example: AES.
- Asymmetric cryptography: Uses a pair of keys: a public key and a private key. Example: RSA.

The main difference is that symmetric cryptography uses one shared secret key, while asymmetric cryptography uses two mathematically related keys.

c. List three common goals of cryptography.

1. Confidentiality: Prevents unauthorized people from reading information.
2. Integrity: Ensures that information has not been modified without authorization.
3. Authentication: Verifies the identity of a user or the origin of a message.